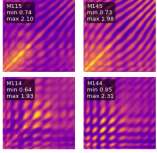


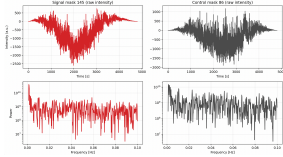
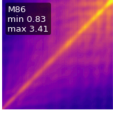
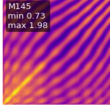
Raw mask time trace and FFT inspection

Friday, January 23, 2026 3:03 pm

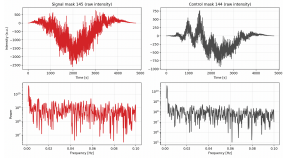
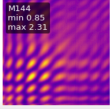
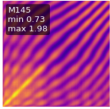


Results for A073

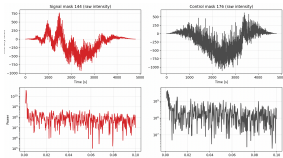
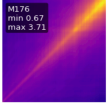
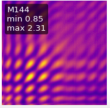
Brage mask 145, control mask 86:



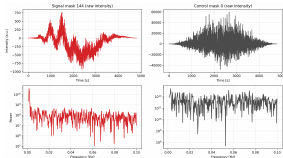
Brage mask 145, control mask 144:



Brage mask 144, control mask 176:



Brage mask 144, control mask 0:



Brage mask 144, control mask 3:



Methodology

1. Raw detector data

You start from the raw detector frames

$$I_{raw}(t, x, y)$$

where

- $t = 0, 1, \dots, T-1$ is the frame index
- (x, y) are detector pixel coordinates.

2. ROI masking

From the dynamic ROI map you construct a binary mask for ROI k :

$$M_k(x, y) = \begin{cases} 1 & \text{if pixel } (x, y) \in \text{ROI } k \\ 0 & \text{otherwise} \end{cases}$$

Only pixels where $M_k=1$ are used.

3. Frame-wise ROI intensity

For each frame, you compute the **mean intensity in the ROI**:

$$I_k(t) = \frac{1}{N_k} \sum_{x,y} M_k(x, y) I_{raw}(t, x, y)$$

where

$$N_k = \sum_{x,y} M_k(x, y)$$

is the number of pixels in the ROI.

This produces a scalar time series:

$$I_k(t), t = 0, \dots, T-1$$

4. Mean removal (DC subtraction)

To isolate **fluctuations**, the long-time mean is removed:

$$\bar{I}_k = \frac{1}{T} \sum_{t=0}^{T-1} I_k(t)$$

$$\delta I_k(t) = I_k(t) - \bar{I}_k$$

This step guarantees:

- the trace is centred around zero
- absolute intensity is removed
- only fluctuations contribute

This is why the trace can start and end near zero even when the raw images are bright.

5. Optional linear detrending (slow drift removal)

A linear model is fit and subtracted:

$$\delta I_k(t) \approx at + b$$

$$\tilde{I}_k(t) = \delta I_k(t) - (at + b)$$

This removes:

- slow intensity ramps
- beam decay / thermal drift
- scan envelope effects

What remains is **stationary fluctuation content**.

6. Windowing (spectral leakage suppression)

Before the FFT, a Hann window is applied:

$$w(t) = \frac{1}{2} \left[1 - \cos \left(\frac{2\pi t}{T-1} \right) \right]$$

$$I_k^{(w)}(t) = \tilde{I}_k(t) w(t)$$

This suppresses artificial frequency leakage caused by finite time boundaries.

Effect:

- The endpoints go smoothly to zero
- Edge discontinuities are removed
- Power from a real frequency stays localized instead of spreading

Why this matters for your data. Your time traces have:

- slow decay envelopes
- long period modulations
- non-integer numbers of cycles

Without a window:

- low-frequency power leaks into neighboring frequencies
- FFTs can look "busy" or artificially broadband
- comparing signal vs control becomes harder

With a Hann window:

- true low-frequency excess stands out more clearly
- the FFT comparison becomes more honest, not less
- you reduce false structure at mid and high frequencies

7. Discrete Fourier Transform

The real FFT is computed:

$$\tilde{I}_k(f) = \sum_{t=0}^{T-1} I_k^{(w)}(t) e^{-2\pi i f t \Delta t}$$

where:

- Delta t is the frame time (seconds)
- frequencies are

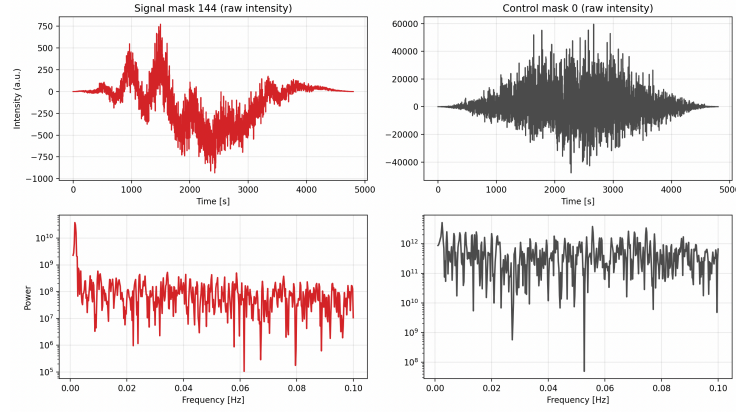
$$f_n = \frac{n}{T \Delta t}, \quad n = 0, 1, \dots$$

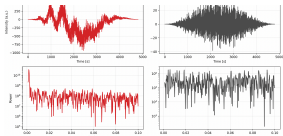
8. Power spectrum

The power spectrum is defined as:

$$P_k(f) = |\tilde{I}_k(f)|^2$$

with the DC component ($f=0$) explicitly ignored.





Interpretation:

Signal ROI (mask 145)

- Huge amplitude relative to control (orders of magnitude larger)
- Very clear low-frequency structure riding under the noise
- Broad "U-shaped" envelope, but with slower modulations on top
- Importantly: the fluctuations are not symmetric noise around zero

This is **not** what random speckle noise looks like.

Control ROI (mask 10)

- Much smaller amplitude
- Symmetric, stationary-looking fluctuations
- No obvious slow modulation, just noise + envelope from beam decay

This looks exactly like what you expect from:

- photon counting noise
- slow beam intensity decay
- no internal dynamics

So already at the time-trace level:

☞ **signal and control are qualitatively different**

2 FFTs (bottom row): the decisive evidence

This is where the story really locks in.

Signal FFT (bottom-left)

- Strong enhancement at very low frequency
- Power rises sharply as $f \rightarrow 0$
- Clear excess spectral weight below ~0.01 Hz

This is exactly what a slow, quasi-periodic modulation produces in FFT space.

Even if you don't see a razor-sharp peak, this is not white noise.

Control FFT (bottom-right)

- Flat-ish spectrum on log scale
- No low-frequency enhancement
- No excess near the signal's low-frequency region

This is basically textbook:

"noise + counting statistics + slow drift removed"

3 The critical comparison

Let me say this very explicitly:

There is no shared spectral feature between signal and control.

That immediately rules out:

- beam current oscillations
- monochromator feedback
- shutter timing artifacts
- detector electronics oscillations
- temperature controller cycling (unless it is highly spatially selective, which is implausible)

If this were instrumental, **both FFTs would light up in the same place.**

They don't.

This is what you plot in log scale.

9. Peak frequency selection

Within a physically motivated band:

$$f_{\min} \leq f \leq f_{\max}$$

you identify the dominant oscillatory component:

$$f_{\text{peak}} \equiv \arg \max_f P_T(f)$$

10. Period extraction

The oscillation period is:

$$T_{\text{peak}} = 1 / f_{\text{peak}}$$

This is the quantity compared across:

- signal ROI
- control ROI
- different diagonal start times
- TTC lineouts

11. Interpretation logic

- Raw images show absolute photon counts
- Mean-subtracted traces show fluctuations
- FFT peaks indicate coherent temporal structure
- Control ROI should show:
 - broadband noise
 - no persistent peak
- Signal ROI shows:
 - a reproducible peak
 - consistent with TTC oscillations

Mean removal and detrending **cannot create oscillations** — they only remove low-frequency content.